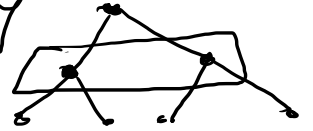


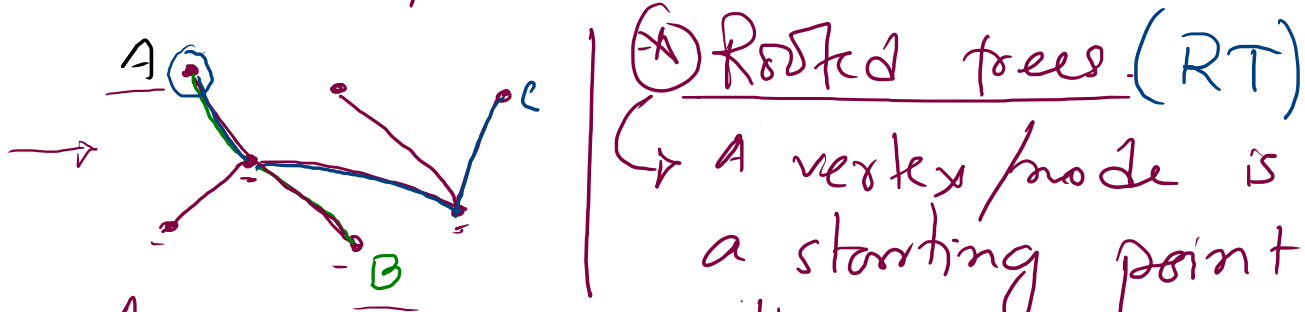
Ch-11.1 Trees.

(*) **Tree** → CG with no simple circuit

$m \leq 2$
binary
 $m=2$

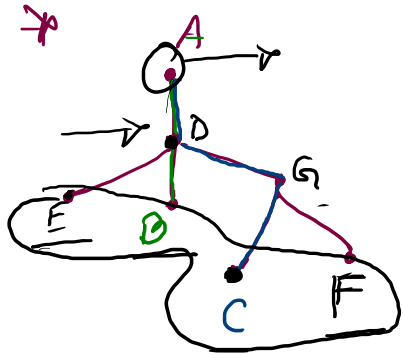


→ Any two vertices has a unique path.



(*) Rooted trees (RT)

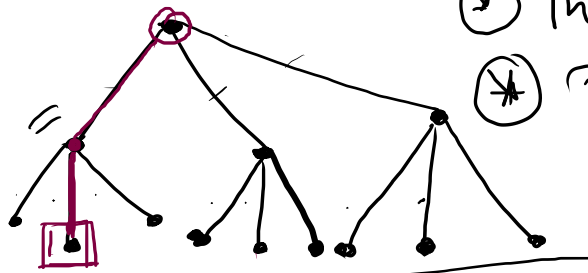
→ A vertex/node is designated as a root/
a starting point and every edge
will span away from the root.



(*) RT → m-ary tree if → every internal vertices
has no more than m children
→ full m-ary tree if → every internal vertices
has exactly m children

Properties of trees.

① $\mathcal{T}_1 \rightarrow$ A tree with n vertices has $(n-1)$ edges



② $\mathcal{T}_2 \rightarrow$ A full m -ary tree with

a/ n - vertices

$$i = (n-1)/m \rightarrow iV$$

$$l = \lceil (m-1)i + 1 \rceil / m \rightarrow \text{leaves.}$$

$$m=3, h=2, l=9=3^2=m^h$$

$$\lceil \log_3 9 \rceil = 2$$

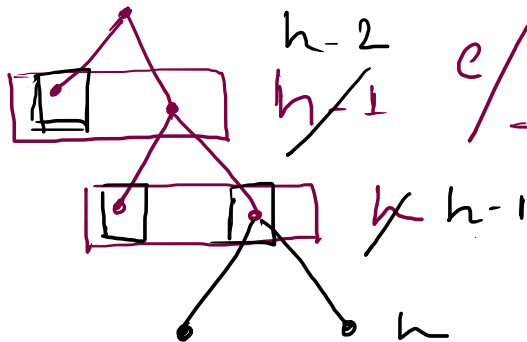
$$n = 13$$

$$l = \lceil n-1 \rceil = 12$$

b/ i internal vertices

$$n = mi + 1 \rightarrow \text{vertices}$$

$$l = (m-1)i + 1 \text{ leaves.}$$



c/ l leaves

$$n = (ml - 1)/(m-1) \rightarrow \text{vertices.}$$

$$i = (l-1)/(m-1) \rightarrow iV$$

⑩ level of a node $\rightarrow$ length of the path to a node from the root.

⑩ height of a tree $\rightarrow \forall n \max(\text{level}_n)$

⑩ root m-ary tree

balanced/unbalanced? $\rightarrow$ if height is (h) and all the leaves are in h or $(h-1)$ levels.

$\rightarrow$ with height h can have at most m^h leaves.

$\rightarrow$ it has height h and l leaves $\rightarrow h \geq \lceil \log_m l \rceil$
but if it is full and balanced $\rightarrow h = \lceil \log_m l \rceil$